

Non-celiac gluten sensitivity. Is it in the gluten or the grain?

Celiac disease is an immune-mediated inflammatory disorder of the small intestine caused by sensitivity to dietary gluten and related proteins in genetically predisposed individuals. Over the past several years, the concept of non-celiac gluten sensitivity (NCGS) has gained significant interest from the scientific community and mass media and the number of individuals embracing a gluten-free diet is rapidly growing. This condition is characterized by gastrointestinal or extraintestinal symptoms that respond to gluten withdrawal without evidence for underlying celiac disease or wheat allergy. Symptoms display significant overlap with the irritable bowel syndrome. Many important factors regarding this relatively novel condition remain to be elucidated; no discriminative markers to support a diagnosis of gluten sensitivity have been identified yet and its pathogenesis remains obscure. Here we review the current knowledge on NCGS, and outline potential pathogenic pathways of different gluten related disorders in order to gain clues about the pathophysiology of this novel condition.